Example consider $\underbrace{-u''}_{Lu} = f$, $u(a) = 0 = u(b)$

Here, $L_h u(x) = -\dfrac{u(x+h) - 2u(x) - u(x-h)}{h^2}$ so that $L_h = \frac{1}{h^2}\begin{pmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & \ddots & \ddots & -1 \\ & & -1 & 2 \end{pmatrix}$, $u_h = \begin{pmatrix} u(a+h) \\ \vdots \\ u(b-h) \end{pmatrix}$

and $f_h = (f(a+h), \dots, f(b-h))^T$.

Consistency? We have to check $L_h R_h u - f_h$ which is here given as

$$L_h R_h u - f_h = L_h R_h u - R_h f = L_h R_h u - R_h \underset{\text{u is exact solution}}{L u} = L_h R_h u - R_h(-\Delta u)$$

$$= \frac{1}{h^2}\begin{pmatrix} 2 & -1 & & \\ -1 & 2 & -1 & \\ & \ddots & \ddots & -1 \\ & & -1 & 2 \end{pmatrix}\begin{pmatrix} u(a+h) \\ \vdots \\ u(b-h) \end{pmatrix} - \begin{pmatrix} u''(a+h) \\ \vdots \\ u''(b-h) \end{pmatrix} \quad \rightsquigarrow \text{Taylor expansion } \mathcal{O}(h^2)$$

$$\Rightarrow \quad \| L_h R_h u - f_h \|_{\infty,h} = \max_{1 \le i \le N-1} \left\| u''(a+ih) - \frac{1}{h^2}\big(u(a+(i+1)h) - 2u(a+ih) + u(a+(i-1)h)\big) \right\|$$

$\in \mathcal{O}(h^2)$ $\rightarrow$ consistent of order 2 w.r.t. the <u>maximum norm</u>! (if $u \in C^4$!)

Lecture 09

11.11.2025

Theorem ("Consistency + stability = convergence")

If the finite difference method $L_h u_h = f_h$ for $Lu = f$ is <u>consistent and stable</u> w.r.t. $(\|\cdot\|_h)_{h>0}$, then it is <u>convergent</u> w.r.t. $(\|\cdot\|_h)_{h>0}$. If it is consistent of order p, then it is convergent of order p.

Proof For invertible L_h (result from the <u>stability</u> assumption), we have $L_h u_h = f_h \Leftrightarrow u_h = L_h^{-1} f_h$.

$$\Rightarrow \|u_h - R_h u\|_h = \|L_h^{-1} f_h - R_h u\|_h = \|L_h^{-1} f_h - L_h^{-1} L_h R_h u\|_h \le \underbrace{|||L_h^{-1}|||_h}_{\substack{\text{stability} \\ \le C < \infty}} \cdot \underbrace{\|f_h - L_h R_h u\|_h}_{\substack{\mathcal{O}(h^p) \text{ due to} \\ \text{consistency}}} \qquad \blacksquare$$

We already know (from Taylor) that our centered finite difference schemes are consistent of order 2. We can therefore expect quadratic convergence <u>if</u> we can show that our method is stable. $\rightsquigarrow$ proof via "M-matrices" following

Definition Let $A \in \mathbb{R}^{d \times d}$. If

a) $a_{ii} > 0$ for all $i = 1, \dots, d$ and $a_{ij} \le 0$ for $i \ne j$

b) $\det(A) \ne 0$

c) $A^{-1} \ge 0$ (componentwise)

then A is called an <u>M-matrix</u>.

Theorem Let $A \in \mathbb{R}^{d\times d}$ be an M-matrix. If there exists $w \in \mathbb{R}^d$ s.t. $\underline{\underline{Aw \geq \mathbb{1}}} = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}$,

then $|||A^{-1}|||_\infty \leq \|w\|_\infty$.

Proof Consider arbitrary $y = (y_1, \dots, y_d)^T \in \mathbb{R}^d$ and define $|y| = (|y_1|, \dots, |y_d|) \in \mathbb{R}^d$.

$$\Rightarrow \quad |y| \overset{\text{elementwise}}{\leq} (\max_i |y_i|, \dots, \max_i |y_i|)^T = \|y\|_\infty \cdot \mathbb{1} \overset{\text{assumption}}{\leq} \|y\|_\infty \cdot Aw \qquad (*)$$

Since A is an M-matrix, we know that $A^{-1} \geq 0$ componentwise so that

$$|A^{-1}y|_i = \left| \sum_{j=1}^d [A^{-1}]_{ij} y_j \right| \leq \sum_{j=1}^d [A^{-1}]_{ij} |y_j| = [A^{-1}|y|]_i \qquad \text{for all } i = 1, \dots, d.$$

$$\Rightarrow |A^{-1}y| \overset{\text{previous ineq comp.wise}}{\leq} A^{-1}|y| \overset{(*)}{\leq} A^{-1}\|y\|_\infty Aw = \|y\|_\infty w \leq \|y\|_\infty (\|w\|_\infty, \|w\|_\infty, \dots, \|w\|_\infty)^T$$

$$\Rightarrow \|A^{-1}y\|_\infty \overset{\text{by def}}{=} \max_{1\leq i\leq d} |A^{-1}y|_i \leq \|w\|_\infty \|y\|_\infty$$

$$\Rightarrow \frac{\|A^{-1}y\|_\infty}{\|y\|_\infty} \leq \|w\|_\infty$$

since this holds for arbitrary y, we conclude that

$$\sup_{\substack{y\in\mathbb{R}^d \\ y\neq 0}} \frac{\|A^{-1}y\|_\infty}{\|y\|_\infty} =: |||A^{-1}||| \leq \|w\|_\infty \qquad \blacksquare$$

Theorem (Stability of the 5-point stencil)

Let $L_h = -1/h^2 \begin{pmatrix} T & I & & \\ I & T & \ddots & \\ & \ddots & \ddots & I \\ & & I & T \end{pmatrix} \in \mathbb{R}^{(N-1)^2 \times (N-1)^2}$, $T = \begin{pmatrix} -4 & 1 & & \\ 1 & -4 & 1 & \\ & \ddots & \ddots & \ddots \\ & & 1 & -4 \end{pmatrix} \in \mathbb{R}^{(N-1)\times(N-1)}$

be the matrix of the 5-point difference stencil. Then:

a) L_h is an M-matrix.

b) $|||L_h^{-1}|||_\infty \leq \frac{1}{8}$ for all $h > 0$ ($\Rightarrow L_h$ is stable!).

Proof

a) We would have to show: i) $(L_h)_{ii} > 0$, $(L_h)_{ij} \leq 0$, $i \neq j$ ii) $\det(L_h) \neq 0$ (see exercises) iii) $L_h^{-1} \overset{\text{comp. wise}}{\geq} 0$ (difficult!)

b) Let us define $w(x_1, x_2) := \frac{x_1 - x_1^2}{2} = \frac{1}{2} x_1 (1 - x_1)$ and $w_h = R_h w$, the restriction of w to Ω_h (recall w_h vectorized version of $w_{k,l} = w(kh, lh)$).

For an interior grid point, we then obtain

$$L_h w_h \overset{\text{row in this vector}}{\rightsquigarrow} -\frac{1}{h^2}(w_{k+1,l} + w_{k,l+1} - 4w_{k,l} + w_{k-1,l} + w_{k,l-1}) \qquad (\text{now use } w_{k,l} = w_{k,l+1} = w_{k,l-1})$$

$$\overset{w \text{ is indep. of } x_2}{=} \frac{1}{h^2}(2w_{k,l} - w_{k-1,l} - w_{k+1,l})$$

$$\overset{\text{use def of } w}{=} \frac{1}{h^2}\left(\cancel{2}\,\frac{\cancel{kh} - k^2h^2}{2} - \frac{(k-\cancel{1})h - (k-1)^2h^2}{2} - \frac{(k+\cancel{1})h - (k+1)^2h^2}{2}\right)$$

$$= \frac{1}{2h^2}\left(-2k^2h^2 + (k-1)^2h^2 + (k+1)^2h^2\right) = -\frac{1}{2}\left(2k^2 - (k-1)^2 - (k+1)^2\right)$$

$$= -\frac{1}{2}\left(2k^2 - (k^2 - 2k + 1) - (k^2 + 2k + 1)\right) = 1$$

We can do similar derivations for nodes on the boundary, hence $L_h w_h \geq 1$.

$$\overset{\text{prev. Thm}}{\Longrightarrow} \quad |||L_h^{-1}|||_\infty \leq \|w_h\|_\infty = \max_{k,l} |w(kh, lh)| \leq \max_{x \in [0,1]} \frac{x - x^2}{2} = \frac{1}{8}. \qquad \blacksquare$$

<u>Corollary</u> Let $-\Delta u = f$ in $\Omega = (0,1)^2$, $u = g$ on $\partial\Omega$ be given. If the exact solution satisfies $u \in C^4(\overline{\Omega})$, then the finite difference method with the 5-point stencil converges of order 2 with respect to the system of maximum norms.